

PRACTICAL LAB :

Create an instance

Connect to AWS Create new machine

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	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	controller1	i-05e5c8d8a35e2c093	Stopped Refresh Refresh	t3.micro	-	View alarms +	ap-south-2a
☐	server1	i-002caa7f2f076c53e	Stopped Refresh Refresh	t3.micro	-	View alarms +	ap-south-2a
☐	docker	i-0933b5d3883ff43f1	Running Refresh Refresh	t3.micro	-	View alarms +	ap-south-2a

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Open browser (get.docker.com)

```
← ↻ https://get.docker.com
# This script is designed to upgrade an existing Docker installation. When using the
# script to update an existing installation, dependencies may not be updated
# to the expected version, resulting in outdated versions.
#
# Source code is available at https://github.com/docker/docker-install/
#
# Usage
# =====
#
# To install the latest stable versions of Docker CLI, Docker Engine, and their
# dependencies:
#
# 1. download the script
#
# $ curl -fsSL https://get.docker.com -o install-docker.sh
#
# 2. verify the script's content
#
# $ cat install-docker.sh
#
# 3. run the script with --dry-run to verify the steps it executes
#
# $ sh install-docker.sh --dry-run
#
# 4. run the script either as root, or using sudo to perform the installation.
#
# $ sudo sh install-docker.sh
#
# Command-line options
# =====
```

Always docker run on root user (#)

sudo su -

```
root@ip-172-31-12-238: ~
ubuntu@ip-172-31-12-238:~$ sudo su -
root@ip-172-31-12-238:~#
```

Download and install docker

```
curl -fsSL https://get.docker.com -o install-docker.sh
```

```
ubuntu@ip-172-31-15-92:~$ sudo su -  
root@ip-172-31-15-92:~# curl -fsSL https://get.docker.com -o install-docker.sh  
root@ip-172-31-15-92:~# |
```

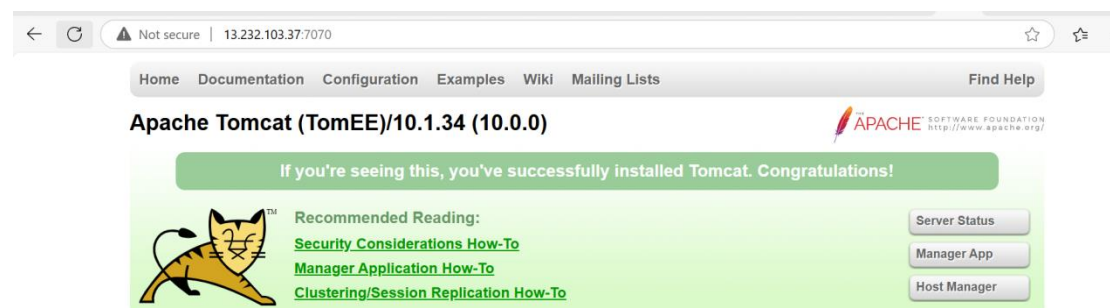
```
sh install-docker.sh
```

```
ubuntu@ip-172-31-12-238:~$ sudo su -  
root@ip-172-31-12-238:~# curl -fsSL https://get.docker.com -o install-docker.sh  
root@ip-172-31-12-238:~# sh install-docker.sh  
# Executing docker install script, commit: 4c94a56999e10efcf48c5b8e3f6afea464f9108e  
+ sh -c apt-get -qq update >/dev/null
```

TOMCAT INSTALLATION:

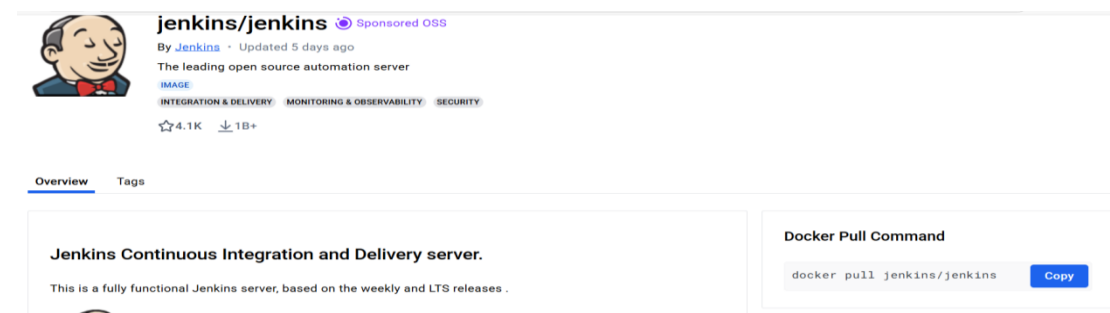
```
#docker run --name mytomcat -p 7070:8080 tomee  
docker run --name mytomcat -p 7070:8080 -d tome
```

Take the public ip and add 7070 to access tomcat



JENKINS INSTALATION:

```
docker run --name jenkins -p 9090:8080 jenkins/jenkins
```



```

root@ip-172-31-12-238:~# docker run --name jenkins -p 9090:8080 jenkins/jenkins
Unable to find image 'jenkins/jenkins:latest' locally
latest: Pulling from jenkins/jenkins
fd0410a2d1ae: Extracting [=====] 42.76MB/48.48MB
007307ec8e93: Download complete
1c6f0cfeaa6c: Download complete
d659880150a8: Download complete
8733ebe6e173: Download complete
32deaa29f9f6: Download complete
0dd998eb9cf5: Download complete
0110ea549d15: Download complete
d5cefc9d910d4: Download complete
cc0276eb9274: Download complete
67c4d9d41045: Download complete
2810efbbb3e: Download complete

```



Using docker file:

Using docker file

This is a **simple text file**, which uses predefined keywords for creating customized docker images.

Key words used in docker file (case sensitive)

- 1) FROM -- used to specify the base image from which the docker file has to be created.
- 2) MAINTAINER -- This represents name of the organization or the author who created this docker file.
- 3) CMD -- This is used to specify the initial command that should be executed when the container starts.
- 4) ENTRYPOINT - used to specify the default process that should be executed when container starts. It can also be used for accepting arguments from the CMD instruction.
- 5) RUN -- Used for running linux commands within the container. It is generally helpful for installing the software in the container.
- 6) USER -- used to specify the default user who should login into the container.

```

3) CMD -- This is used to specify the initial command that should be executed when the container starts.

4) ENTRYPOINT - used to specify the default process that should be executed when container starts. It can also be used for accepting arguments from the CMD instruction.

5) RUN -- Used for running linux commands within the container. It is generally helpful for installing the software in the container.

6) USER -- used to specify the default user who should login into the container.

7) WORKDIR -- Used to specify default working directory in the container

8) COPY -- Copying the files from the host machine to the container.

9) ADD -- Used for copying files from host to container, it can also be used for downloading files from remote servers.

```

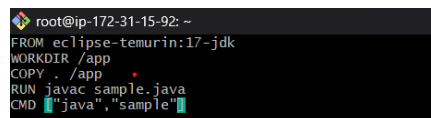
Run: is used to execute commands while building the Docker image.
Cmd: defines the default command **that runs** when the container starts.

Create dockerfile: **vim dockerfile**

```

FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY . /app
RUN javac sample.java
CMD ["java","sample"]

```



```

root@ip-172-31-15-92: ~
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY . /app
RUN javac sample.java
CMD ["java","sample"]

```

:wq

Create sample.java(calculator program)

vim sample.java

```

import java.util.Scanner;

class sample{
    public static void main(String[] args) {

        char operator;
        Double number1, number2, result;

        // create an object of Scanner class
        Scanner input = new Scanner(System.in);

        // ask users to enter operator
        System.out.println("Choose an operator: +, -, *, or /");
        operator = input.next().charAt(0);

        // ask users to enter numbers
        System.out.println("Enter first number");
        number1 = input.nextDouble();

        System.out.println("Enter second number");
        number2 = input.nextDouble();

        switch (operator) {

```

```

// performs addition between numbers
case '+':
    result = number1 + number2;
    System.out.println(number1 + " + " + number2 + " = " + result);
    break;

// performs subtraction between numbers
case '-':
    result = number1 - number2;
    System.out.println(number1 + " - " + number2 + " = " + result);
    break;

// performs multiplication between numbers
case '*':
    result = number1 * number2;
    System.out.println(number1 + " * " + number2 + " = " + result);
    break;

// performs division between numbers
case '/':
    result = number1 / number2;
    System.out.println(number1 + " / " + number2 + " = " + result);
    break;

default:
    System.out.println("Invalid operator!");
    break;
}

input.close();
}
}

```

Install java

`apt-get install openjdk-17-jdk -y`

compile : `javac sample.java`

run : `java sample`(sample is the file name)

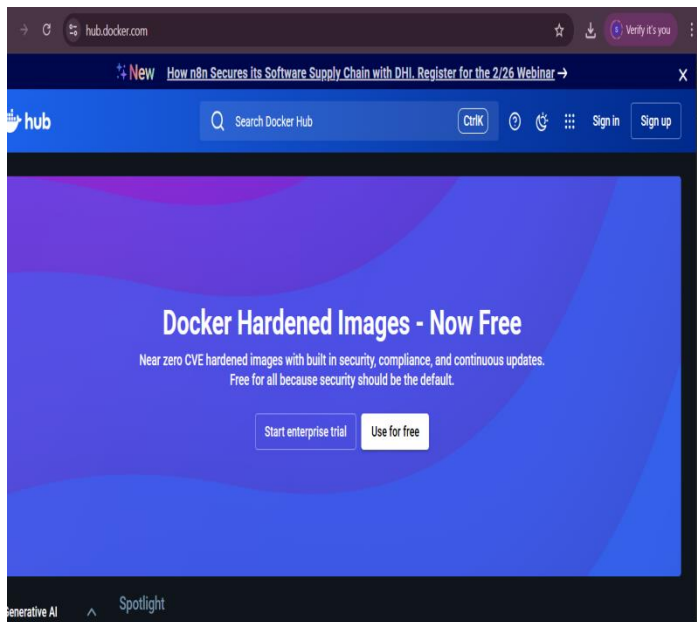
Compile and run the java program

```

root@ip-172-31-12-238:~# javac sample.java
root@ip-172-31-12-238:~# java sample
Choose an operator: +, -, *, or /
+
Enter first number
10
Enter second number
20
10.0 + 20.0 = 30.0
root@ip-172-31-12-238:~#

```

Create a dockerhub account <https://hub.docker.com>



Create a docker image for the java application

docker build -t maheedhar45/javacalculator .

```
root@ip-172-31-12-238:~# docker build -t maheedhar45/javacalculator .
[+] Building 16.8s (9/9) FINISHED
=> [internal] load build definition from dockerfile
=> transferring dockerfile: 119B
=> [internal] load metadata for docker.io/library/openjdk:latest
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/4] FROM docker.io/library/openjdk:latest@sha256:9b448de897d211c9a0ec635a485650aed6e28d4ecalefbc34940560a480b3f1f
=> resolve docker.io/library/openjdk:latest@sha256:9b448de897d211c9a0ec635a485650aed6e28d4ecalefbc34940560a480b3f1f
=> sha256:9b448de897d211c9a0ec635a485650aed6e28d4ecalefbc34940560a480b3f1f 1.04kB / 1.04kB
=> sha256:fe05457a5e9b9403f8e72eaba507ae80a4237d2d3f219fa62ceb128482a9ee 954B / 954B
=> sha256:71260f256d19f4ae5c762601a5301418d2516ca591103b1376f063ba0b7ba056 4.46kB / 4.46kB
=> sha256:197c1adcd755131915cd019bdd58658d44445b3638f65449932c18ee39b6047c 44.56MB / 44.56MB
=> sha256:57b698b7af4b18900b53c768746b1dfb603dfb9aec1eea328fdac86d37001e2a 12.26MB / 12.26MB
=> sha256:95a27dbe0150755fca4304b4af0d7d6dd6a40ede6fdb30da8568e9a8cdf23a9 188.74MB / 188.74MB
=> extracting sha256:197c1adcd755131915cd019bdd58658d44445b3638f65449932c18ee39b6047c
=> extracting sha256:57b698b7af4b18900b53c768746b1dfb603dfb9aec1eea328fdac86d37001e2a
=> extracting sha256:95a27dbe0150755fca4304b4af0d7d6dd6a40ede6fdb30da8568e9a8cdf23a9
=> [internal] load build context
=> transferring context: 33.36kB
=> [2/4] WORKDIR /app
=> [3/4] COPY . /app
=> [4/4] RUN javac sample.java
=> exporting to image
=> exporting layers
=> writing image sha256:4c56e7c240a98376e68ab0d4657190e439631dda16cd5d995f708b51361a29f0
=> naming to docker.io/maheedhar45/javacalculator
root@ip-172-31-12-238:~#
```

docker image ls

```
root@ip-172-31-12-238:~# docker image ls
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
maheedhar45/javacalculator latest      4c56e7c240a9 53 seconds ago 470MB
mycontainer          latest      d88c66989850 18 minutes ago 214MB
jenkins/jenkins      latest      f26f1a7cbb33 5 days ago    466MB
tomee                latest      5567c8574db7 5 weeks ago   367MB
ubuntu               latest      b1d9df8ab815 2 months ago  78.1MB
root@ip-172-31-12-238:~#
```

Now u push the docker image into docker hub

So create dockerhub account (hub.docker.com) (maheedhar@cmritonline.ac.in) (Welcome@1)
Login

docker login -u maheedhar45(username)

Password: Welcome@1

```

root@ip-172-31-12-238:~# docker login -u maheedhar45
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credential-stores

Login Succeeded
root@ip-172-31-12-238:~#

```

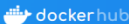
docker push maheedhar45/javacalculator

```

root@ip-172-31-12-238:~# docker push maheedhar45/javacalculator
Using default tag: latest
The push refers to repository [docker.io/maheedhar45/javacalculator]
160ccfe884b2: Pushed
09f939878b71: Pushed
4fc40ef6f284: Pushed
56285d9a7760: Mounted from library/openjdk
077bfff59ce57: Mounted from library/openjdk
9cd9df9ffc97: Mounted from library/openjdk
latest: digest: sha256:c2507d7ca91aa526ef92f9af23afc06ba896f8733269702e0b4af25fc074ca7c size: 1577
root@ip-172-31-12-238:~#

```

Now go and check in the docker hub

 Explore Repositories Organizations Usage Search Docker Hub ctrl+K ? ⚙️ 🔔 🔄 ☰ M				
maheedhar45 Search by repository name All content Create a repository				
Name	Last Pushed	Contains	Visibility	Scout
maheedhar45/javacalculator	in 1 minute	IMAGE	Public	Inactive

Accessing the image

docker run --name myjava -it maheedhar45/javacalculator

```

root@ip-172-31-12-238:~# docker run --name myjava -it maheedhar45/javacalculator
Choose an operator: +, -, *, or /
+
Enter first number
10
Enter second number
20
10.0 + 20.0 = 30.0
root@ip-172-31-12-238:~#

```